

Conditional Flow Matching / Score-based Data Assimilation

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<https://cia-oceanix.github.io/>



Score-based Data Assimilation

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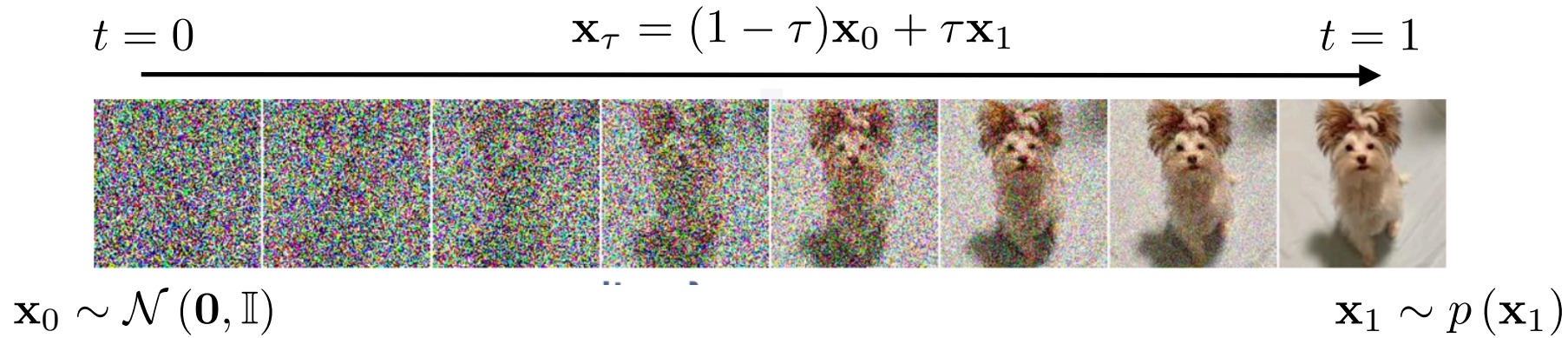
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Abstract

Data assimilation, in its most comprehensive form, addresses the Bayesian inverse problem of identifying plausible state trajectories that explain noisy or incomplete observations of stochastic dynamical systems. Various approaches have been proposed to solve this problem, including particle-based and variational methods. However, most algorithms depend on the transition dynamics for inference, which becomes intractable for long time horizons or for high-dimensional systems with complex dynamics, such as oceans or atmospheres. In this work, we introduce score-based data assimilation for trajectory inference. We learn a score-based generative model of state trajectories based on the key insight that the score of an arbitrarily long trajectory can be decomposed into a series of scores over short segments. After training, inference is carried out using the score model, in a non-autoregressive manner by generating all states simultaneously. Quite distinctively, we decouple the observation model from the training procedure and use it only at inference to guide the generative process, which enables a wide range of zero-shot observation scenarios. We present theoretical and empirical evidence supporting the effectiveness of our method.

Flow matching summary

Flow matching path from a Gaussian space to the targeted physical space

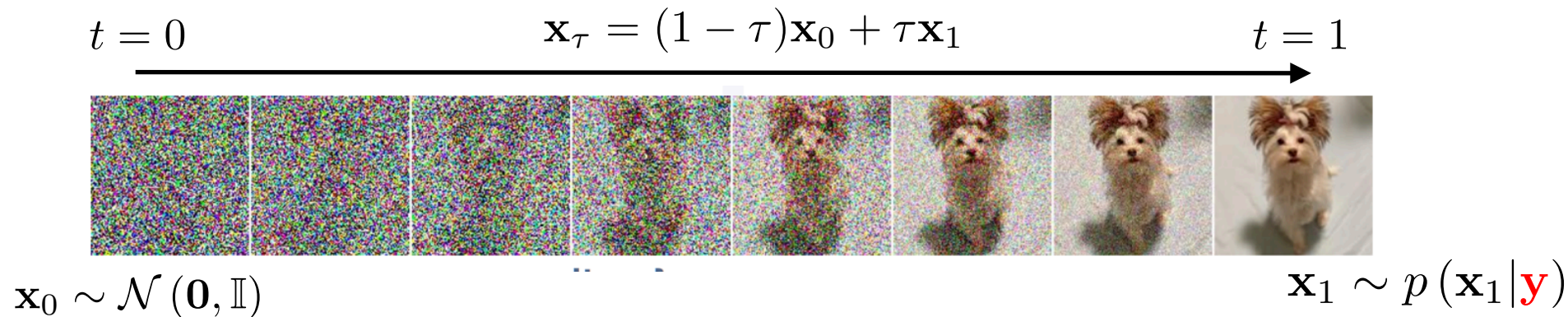


Associated flow-based sampling ODE: $\frac{d\mathbf{x}_\tau}{d\tau} = \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 | \mathbf{x}_\tau]$

Unets as state-of-the-art approximators: $\mathcal{U}_{net}(\mathbf{x}_\tau, \tau) = \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 | \mathbf{x}_\tau]$

Conditional Flow matching summary

Flow matching path from a Gaussian space to the targeted physical space

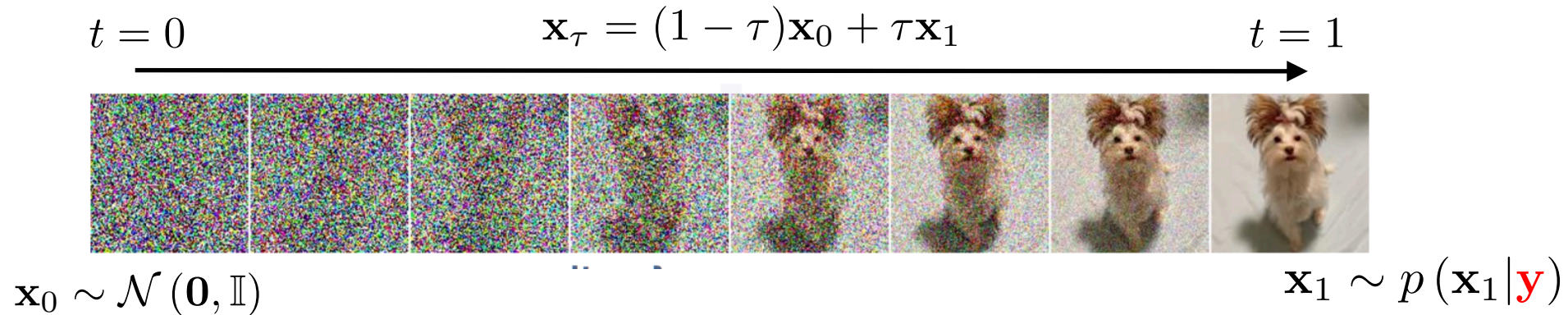


Associated flow-based sampling ODE: $\frac{d\mathbf{x}_\tau}{dt} = \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 | \mathbf{x}_\tau, \mathbf{y}]$

Unets as state-of-the-art approximators: $\mathcal{U}_{net}(\mathbf{x}_\tau, \mathbf{y}, \tau) \approx \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 | \mathbf{x}_\tau, \mathbf{y}]$

Conditional Flow matching using a pre-trained non-conditional model (Adapted from Rozet and Louppe, Neurips'2023)

Flow matching path from a Gaussian space to the targeted physical space



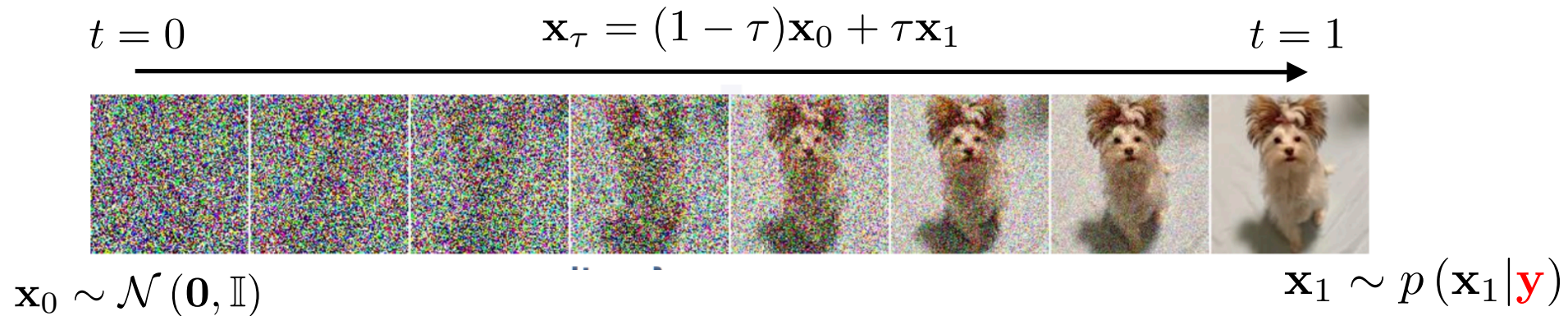
Score-based sampling ODE:
$$\frac{d\mathbf{x}_\tau}{dt} = \frac{1}{1-t}\mathbf{x}_\tau + \frac{1-2t}{(1-t)^2} \nabla_{\mathbf{x}_\tau} \log p_\tau [\mathbf{x}_\tau | \mathbf{y}]$$

$$\nabla_{\mathbf{x}_\tau} \log p_\tau [\mathbf{y} | \mathbf{x}_\tau] + \nabla_{\mathbf{x}_\tau} \log p_\tau [\mathbf{x}_\tau]$$

Bayes rule

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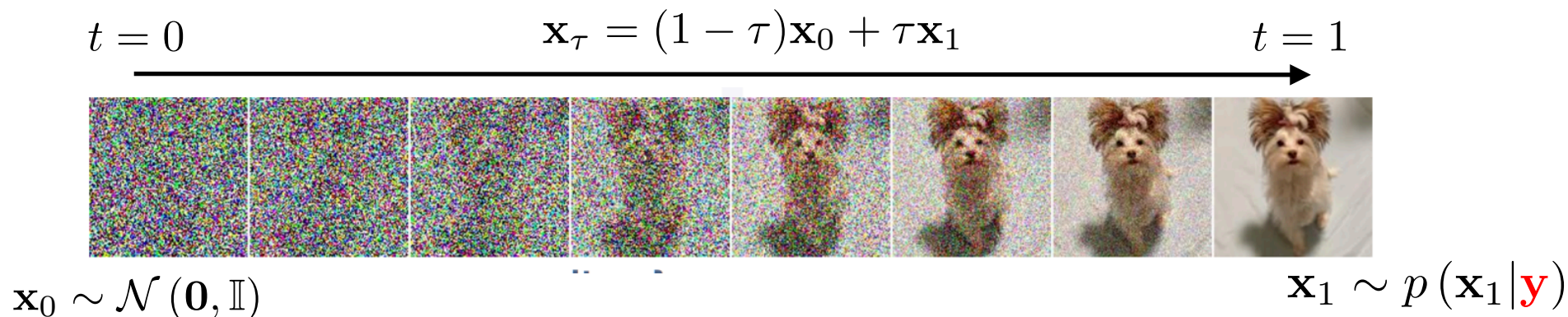
Score-based sampling ODE:
$$\frac{d\mathbf{x}_\tau}{dt} = \frac{1}{1-t}\mathbf{x}_\tau + \frac{1-2t}{(1-t)^2} (\nabla_{\mathbf{x}_\tau} \log p_\tau [\mathbf{y} | \mathbf{x}_\tau] + \nabla_{\mathbf{x}_\tau} \log p_\tau [\mathbf{x}_\tau])$$

$$p_\tau [\mathbf{y} | \mathbf{x}_\tau] \approx \mathcal{G}(\mathbf{y}; \mathbf{H}(\mathbb{E}[\mathbf{x}_1 | \mathbf{x}_\tau]), \Sigma)$$

$$\text{s.t.} \quad \mathbf{y} = \mathbf{H}(\mathbf{x}_1) + \epsilon$$

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Conditional ODE:

$$\frac{d\mathbf{x}_\tau}{dt} = \frac{1}{1-t}\mathbf{x}_\tau + \frac{2t-1}{1-t}\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_\tau, \mathbf{y}]$$

ODE with approximation for the observation term:

$$\frac{d\mathbf{x}_\tau}{dt} \approx \frac{1}{1-t}\mathbf{x}_\tau + \frac{2t-1}{1-t}\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_\tau] + \frac{1-2t}{(1-t)^2}\nabla_{\mathbf{x}_\tau} \log \mathcal{G}(\mathbf{y}; \mathbf{H}(\mathbb{E}[\mathbf{x}_1 | \mathbf{x}_\tau]), \Sigma)$$

References

Rozet and Loupe. Score-based Data Assimilation, Neurips'2023. <https://arxiv.org/abs/2306.10574>

Code: <https://github.com/francois-rozet/sda>